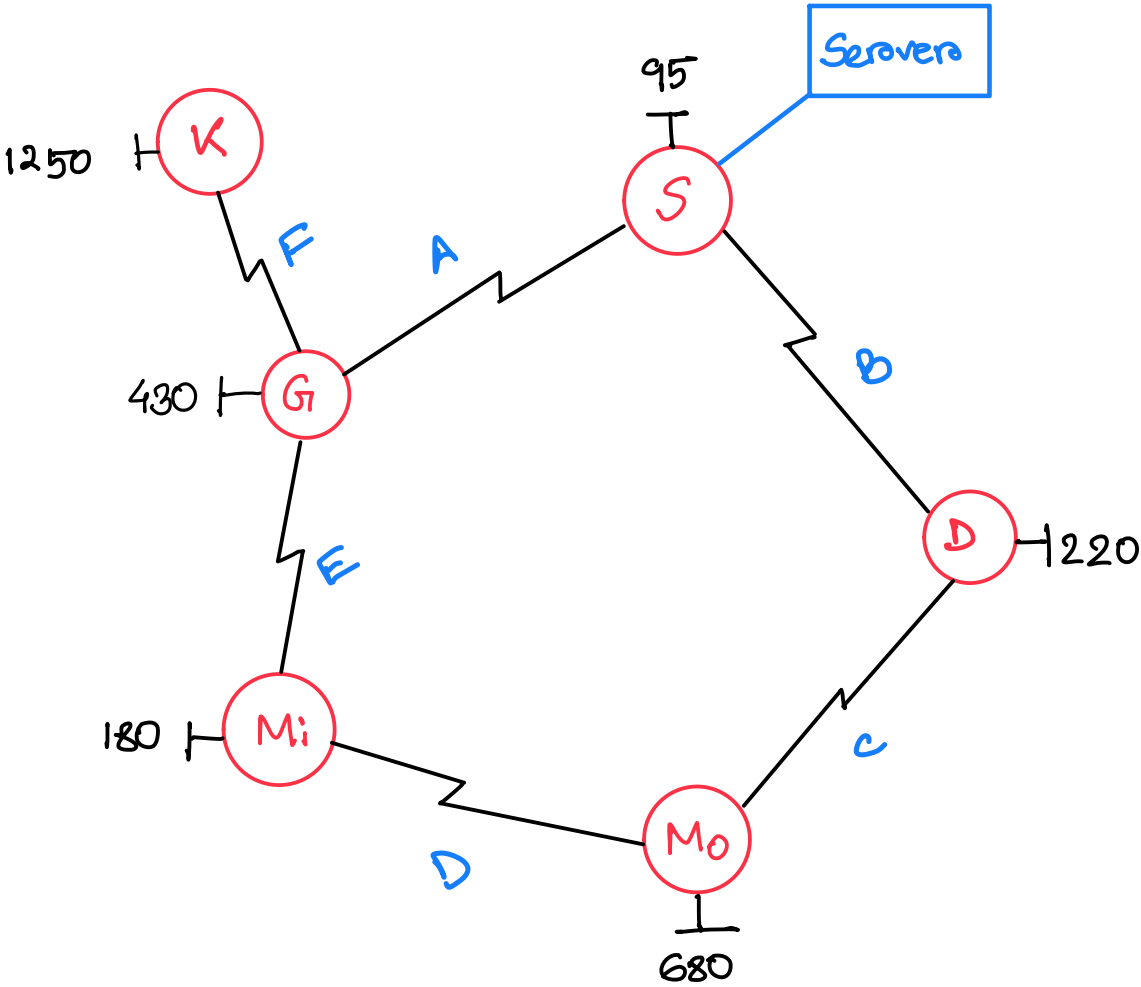


Topology:



VLSM Table:

11	10	9	8	7	6	5	4	3	2	1	0
2048	1024	512	256	128	64	32	16	8	4	2	1

Name	Hosts	Hosts + 2	Block size	Host bits	Prefix
K	1250	1252	2048	11	21
M0	680	682	1024	10	22
G	430	432	512	9	23
D	220	222	256	8	24
Mi	180	182	256	8	24
S	96	98	128	7	25
A	2	4	4	2	30
B	2	4	4	2	30
C	2	4	4	2	30
D	2	4	4	2	30
E	2	4	4	2	30
F	2	4	4	2	30

Network address is $10.3.\bar{0}.\bar{0}/16$

$10.3.0000|0000.0000\ 0000 \rightarrow 0\ K$

$.0000\ 1|000.0000\ 0000 \rightarrow 8.0\ Mb$

$.0000\ 11|00.0000\ 0000 \rightarrow 12.0\ G$

$.0000\ 111|0.0000\ 0000 \rightarrow 14.0\ D$

$.0000\ 1111.0000\ 0000 \rightarrow 15.0\ Mi$

$.00010000|.0000\ 0000 \rightarrow 16.0\ S$

$.0001\ 0000.1|000\ 00|00 \rightarrow 16.128\ A$

$.0001\ 0000.1000\ 01|00 \rightarrow 16.132\ B$

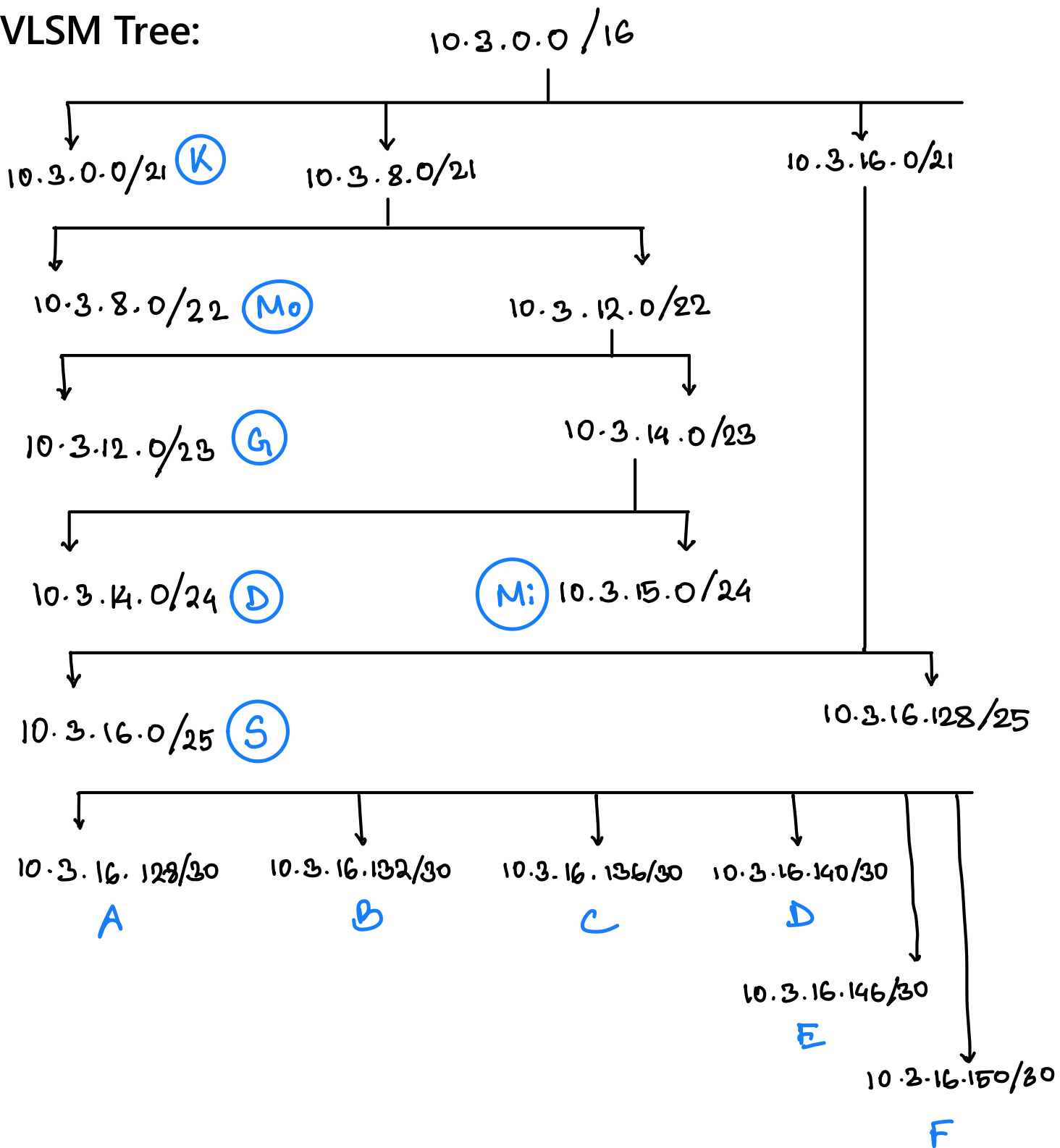
$.0001\ 0000.1000\ 10|00 \rightarrow 16.136\ C$

$.0001\ 0000.1000\ 11|00 \rightarrow 16.140\ D$

$.0001\ 0000.1001\ 00|00 \rightarrow 16.146\ E$

$.0001\ 0000.1001\ 01|00 \rightarrow 16.150\ F$

VLSM Tree:



IP Table:

Area Name	Required Host	Block Size	Mask	Network Address	Usable IP range	Broadcast address
Kamrangirchar (K)	1250	2048	/21	10.3.0.0	10.3.0.1 – 10.3.7.254	10.3.7.255
Motijheel (Mo)	680	1024	/22	10.3.8.0	10.3.8.1 – 10.3.11.254	10.3.11.255
Gulshan (G)	430	512	/23	10.3.12.0	10.3.12.1 – 10.3.13.254	10.3.13.255
Dhanmondi (D)	220	256	/24	10.3.14.0	10.3.14.1 – 10.3.14.254	10.3.14.255
Mirpur (Mi)	180	256	/24	10.3.15.0	10.3.15.1 – 10.3.15.254	10.3.15.255
Secretariat (S)	95	128	/25	10.3.16.0	10.3.16.1 – 10.3.16.126	10.3.16.127
Link A (S ↔ G)	2	4	/30	10.3.16.128	10.3.16.129–10.3.16.130	10.3.16.131
Link B (S ↔ D)	2	4	/30	10.3.16.132	10.3.16.133–10.3.16.134	10.3.16.135
Link C (D ↔ Mo)	2	4	/30	10.3.16.136	10.3.16.137–10.3.16.138	10.3.16.139
Link D (Mo ↔ Mi)	2	4	/30	10.3.16.140	10.3.16.141–10.3.16.144	10.3.16.145
Link E (Mi ↔ G)	2	4	/30	10.3.16.146	10.3.16.147–10.3.16.148	10.3.16.149
Link F (G ↔ K)	2	4	/30	10.3.16.150	10.3.16.151–10.3.16.152	10.3.16.153

Commands:

#####

IP Config

#####

#Secretariat

Router#en

Router#conf t

Router(config)#hostname Secretariat

Secretariat(config)#no ip domain-lookup

Secretariat(config)#interface gigabit0/0

Secretariat(config-if)#ip address 10.3.16.1 255.255.255.128

Secretariat(config-if)#no shutdown

#Dhanmondi

Router>en

Router#conf t

Router(config)#hostname Dhanmondi

Dhanmondi(config)#no ip domain-lookup

Dhanmondi(config)#interface gigabit0/0

Dhanmondi(config-if)#ip address 10.3.14.1 255.255.255.0

Dhanmondi(config-if)#no shutdown

#Motijheel

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Motijheel

Motijheel(config)#no ip domain-lookup

Motijheel(config)#interface gigabit0/0

Motijheel(config-if)#ip address 10.3.8.1 255.255.252.0

Motijheel(config-if)#no shutdown

#Mirpur

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Mirpur

Mirpur(config)#no ip domain-lookup

Mirpur(config)#interface gigabit0/0

Mirpur(config-if)#ip address 10.3.15.1 255.255.255.0

Mirpur(config-if)#no shutdown

#Gulshan

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Gulshan

Gulshan(config)#no ip domain-lookup

Gulshan(config)#interface gigabit0/0

Gulshan(config-if)#ip address 10.3.12.1 255.255.254.0

Gulshan(config-if)#no shutdown

#Kamrangirchar

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Kamrangirchar

Kamrangirchar(config)#no ip domain-lookup

Kamrangirchar(config)#interface gigabit0/0

Kamrangirchar(config-if)#ip address 10.3.0.1 255.255.248.0

Kamrangirchar(config-if)#no shutdown

#####

Serial bus

#####

SECRETARIAT

Secretariat>en

Secretariat#conf t

Secretariat(config)#int se0/0/1

Secretariat(config-if)#ip address 10.3.16.130 255.255.255.252

Secretariat(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down

Secretariat(config-if)#no shutdown

Secretariat(config-if)#clock rate 64000

Secretariat(config-if)#

Secretariat(config-if)#exit

Secretariat(config)#int se0/0/0

Secretariat(config-if)#ip address 10.3.16.133 255.255.255.252

Secretariat(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down

Secretariat(config-if)#clock rate 64000

Secretariat(config-if)#end

GULSHAN

Gulshan>en

Gulshan#conf t

Gulshan(config)#int se0/0/0

Gulshan(config-if)#ip address 10.3.16.129 255.255.255.252

Gulshan(config-if)#no shutdown

Gulshan(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
exit

Gulshan(config)#int se0/1/0

Gulshan(config-if)#ip address 10.3.16.146 255.255.255.252

Gulshan(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down

Gulshan(config-if)#clock rate 64000

Gulshan(config-if)#exit

Gulshan(config)#int se0/0/1

Gulshan(config-if)#ip address 10.3.16.149 255.255.255.252

Gulshan(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down

Gulshan(config-if)#clock rate 64000

Gulshan(config-if)#end

DHANMONDI

Dhanmondi>en

Dhanmondi#conf t

Dhanmondi(config)#int se0/0/0

Dhanmondi(config-if)#ip address 10.3.16.134 255.255.255.252

Dhanmondi(config-if)#no shutdown

Dhanmondi(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit

Dhanmondi(config)#int se0/0/1

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to up

Dhanmondi(config-if)#ip address 10.3.16.137 255.255.255.252

Dhanmondi(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down

Dhanmondi(config-if)#clock rate 64000

Dhanmondi(config-if)#exit

Dhanmondi(config)#ip helper-address 10.3.16.2

MOTIJHEEL

Motijheel>en

Motijheel#conf t

Motijheel(config)#int se0/0/1

Motijheel(config-if)#ip address 10.3.16.138 255.255.255.252

Motijheel(config-if)#no shutdown

Motijheel(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
exit

Motijheel(config)#int se0/0/0

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1,
changed state to up

Motijheel(config-if)#ip address 10.3.16.141 255.255.255.252

Motijheel(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down

Motijheel(config-if)#end

MIRPUR

Mirpur>en

Mirpur#conf t

Mirpur(config)#int se0/0/0

Mirpur(config-if)#ip address 10.3.16.145 255.255.255.252

Mirpur(config-if)#no shutdown

Mirpur(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit

Mirpur(config)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to up

int se0/0/1

Mirpur(config-if)#ip address 10.3.16.142 255.255.255.252

Mirpur(config-if)#no shutdown

Mirpur(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1,
changed state to up

clock rate 64000

Mirpur(config-if)#exit

Mirpur(config)#ip helper-address 10.3.16.2

KAMRANGIRCHAR

Kamrangirchar>en

Kamrangirchar#conf t

Kamrangirchar(config)#int se0/0/0

Kamrangirchar(config-if)#ip address 10.3.16.150 255.255.255.252

Kamrangirchar(config-if)#no shutdown

Kamrangirchar(config-if)#

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit

Kamrangirchar(config)#end

#####

Routing

#####

#Secretariat routing to Gulshan (recursive static route); rest RIP V2

secretariat>en

secretariat#conf t

Enter configuration commands, one per line. End with CNTL/Z.

secretariat(config)#ip route 10.3.12.0 255.255.254.0 10.3.16.129

secretariat(config)#router rip

secretariat(config-router)#version 2

secretariat(config-router)#no auto-summary

secretariat(config-router)#network 10.3.0.0

secretariat(config-router)#end

#Gulshan routing to Kamrangirchar (directly attached static route);
to Gulshan (recursive static route); rest RIP V2

Gulshan>en

Gulshan#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Gulshan(config)#ip route 10.3.0.0 255.255.248.0 serial0/0/1

%Default route without gateway, if not a point-to-point interface,
may impact performance

Gulshan(config)#ip route 10.3.16.0 255.255.255.128 10.3.16.130

Gulshan(config)#router rip

Gulshan(config-router)#version 2

Gulshan(config-router)#no auto-summary

Gulshan(config-router)#network 10.3.0.0

Gulshan(config-router)#end

#Kamrangirchar routing to Gulshan (recursive static route); rest RIP V2

Kamrangirchar>en

Kamrangirchar#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Kamrangirchar(config)#ip route 10.3.12.0 255.255.255.252 serial0/0/0

%Default route without gateway, if not a point-to-point interface,
may impact performance

Kamrangirchar(config)#router rip

Kamrangirchar(config-router)#version 2

Kamrangirchar(config-router)#no auto-summary

Kamrangirchar(config-router)#network 10.3.0.0

Kamrangirchar(config-router)#end

#Mirpur RIP V2

Mirpur>en

Mirpur#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Mirpur(config)#router rip

Mirpur(config-router)#version 2

Mirpur(config-router)#no auto-summary

Mirpur(config-router)#network 10.3.0.0

Mirpur(config-router)#end

#Motijheel to Dhanmondi (floating), RIP for else

Motijheel>en

Motijheel#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Motijheel(config)#ip route 10.3.16.0 255.255.255.128 10.3.16.137 5

Motijheel(config)#router rip

Motijheel(config-router)#version 2

Motijheel(config-router)#no auto-summary

Motijheel(config-router)#network 10.3.0.0

Motijheel(config-router)#end

#Dhanmondi return route for static and RIP v2

Dhanmondi>en

Dhanmondi#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Dhanmondi(config)#ip route 10.3.8.0 255.255.252.0 10.3.16.138

Dhanmondi(config)#router rip

Dhanmondi(config-router)#version 2

Dhanmondi(config-router)#no auto-summary

Dhanmondi(config-router)#network 10.3.0.0

Dhanmondi(config-router)#end